



IQ-TREE and GRASP

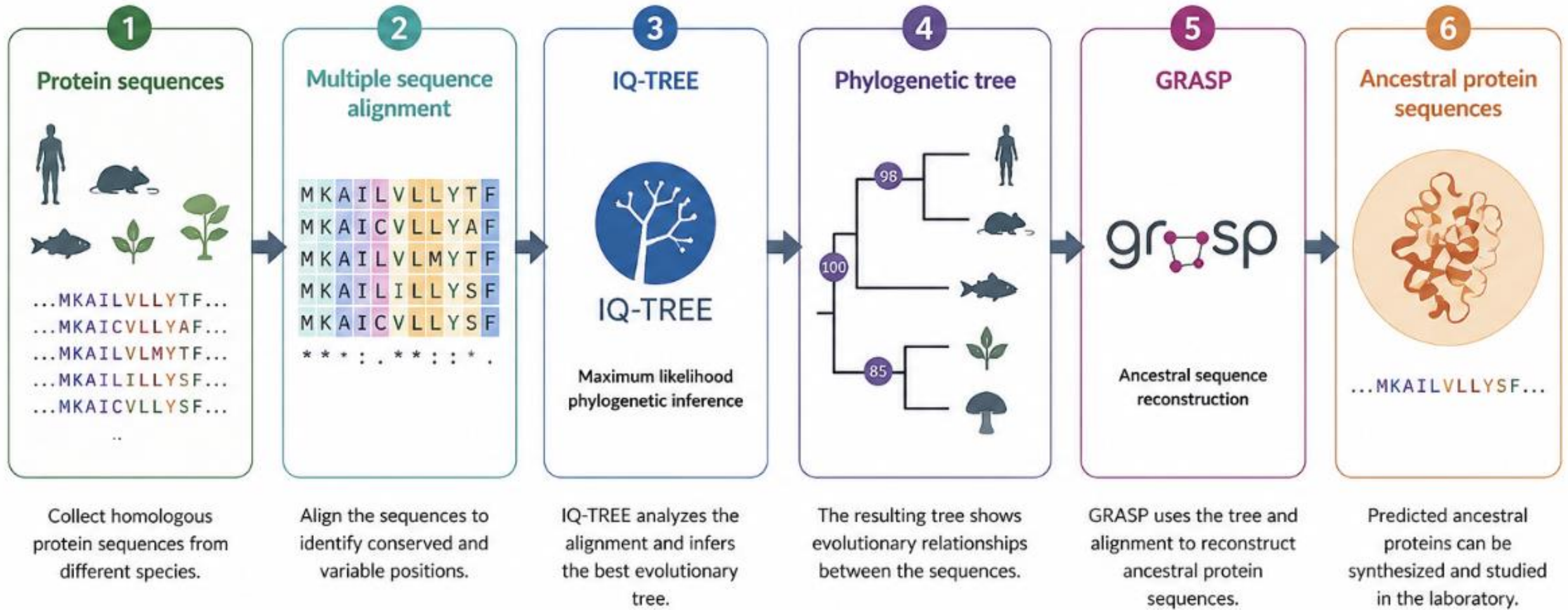
Phylogenetic Interference and
Ancestral Sequence
Reconstruction

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General Workflow



What is IQ-TREE?

IQ-TREE is an open-source software for phylogenetic tree inference using the **Maximum Likelihood (ML)** approach.



FAST

Highly efficient algorithms enable rapid analysis of large datasets.



ACCURATE

Implements advanced models of sequence evolution and model selection (ModelFinder).



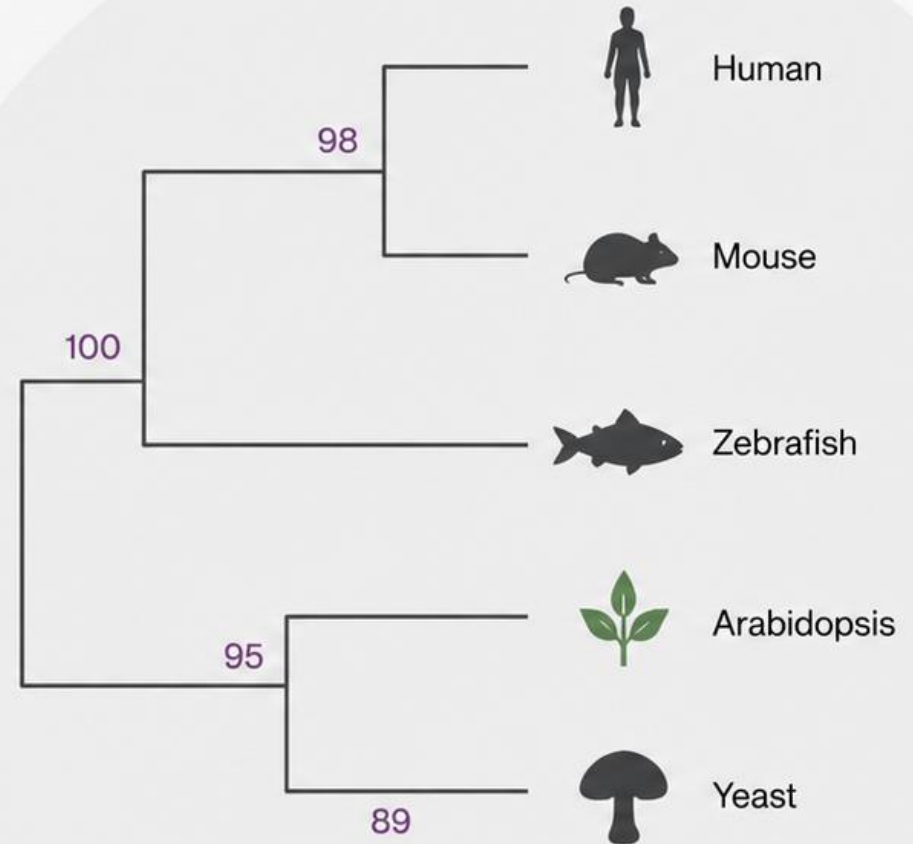
ROBUST

Ultrafast bootstrap provides reliable support values in minimal time.



VERSATILE

Supports DNA, RNA, protein, codon and morphological data with >200 models.



Results from IQ-TREE

The aligned UndB dataset was analyzed using **IQ-TREE** to infer a maximum-likelihood phylogenetic tree.



DATASET

- 124 protein sequences
- 525 amino acid positions



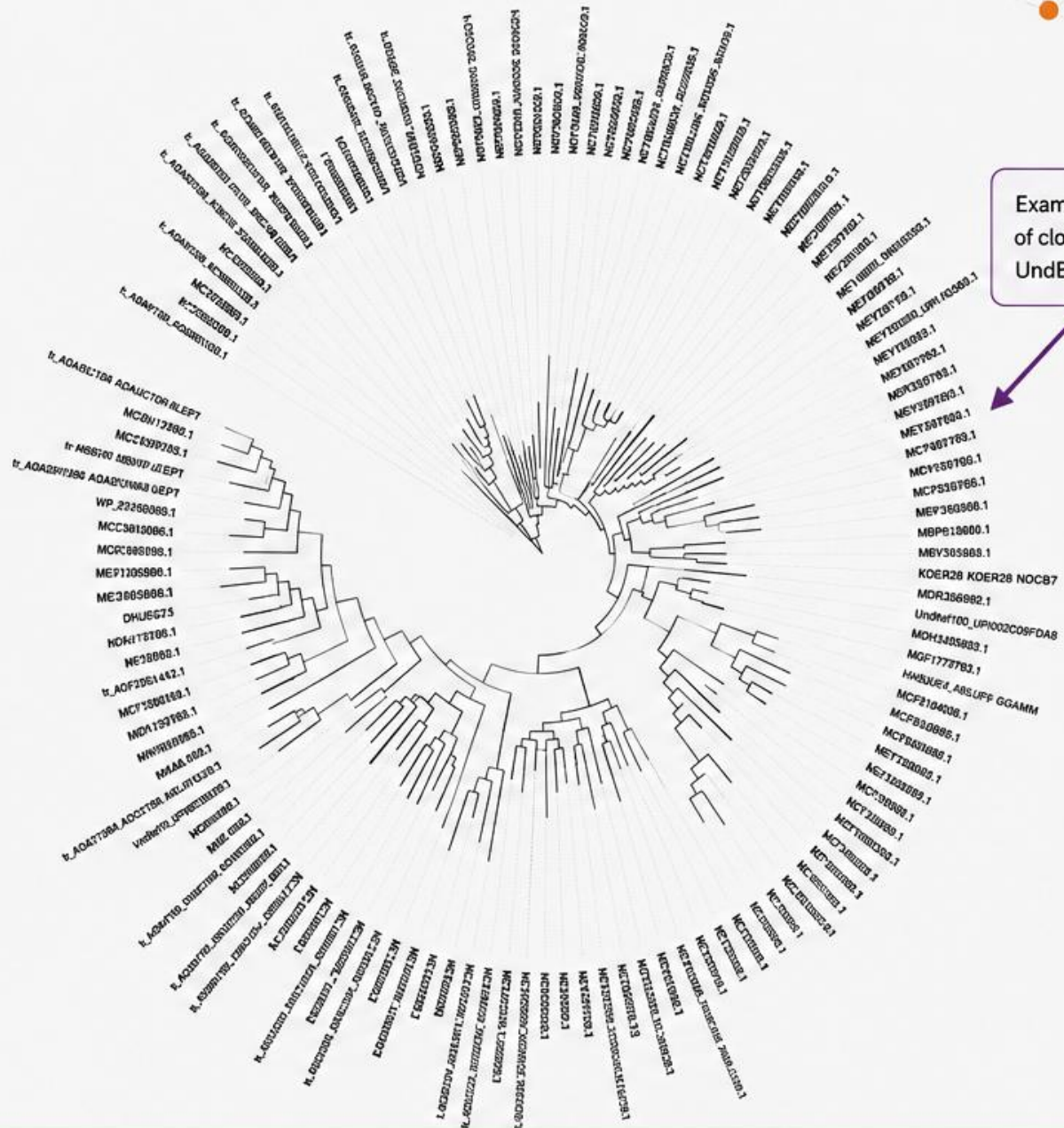
IQ-TREE ANALYSIS

- Maximum Likelihood phylogenetic inference
- ModelFinder selected the best-fitting model: **Q.PFAM+I+R7**
- Tree reconstructed from aligned UndB homologs



PURPOSE

- To determine the evolutionary relationships between UndB proteins from different organisms.



KEY RESULT

IQ-TREE reconstructed the evolutionary relationships among 124 UndB homologs and selected **Q.PFAM+I+R7** as the best fitting

What is GRASP?

GRASP (Graphical Representation of Ancestral Sequence Predictions) reconstructs ancestral protein sequences using phylogenetic trees and sequence alignments.



ANCESTRAL RECONSTRUCTION

Infers ancestral protein sequences at internal nodes of a phylogenetic tree.



HANDLES INSERTIONS & DELETIONS

Accurately models indels, improving ancestral sequence predictions.



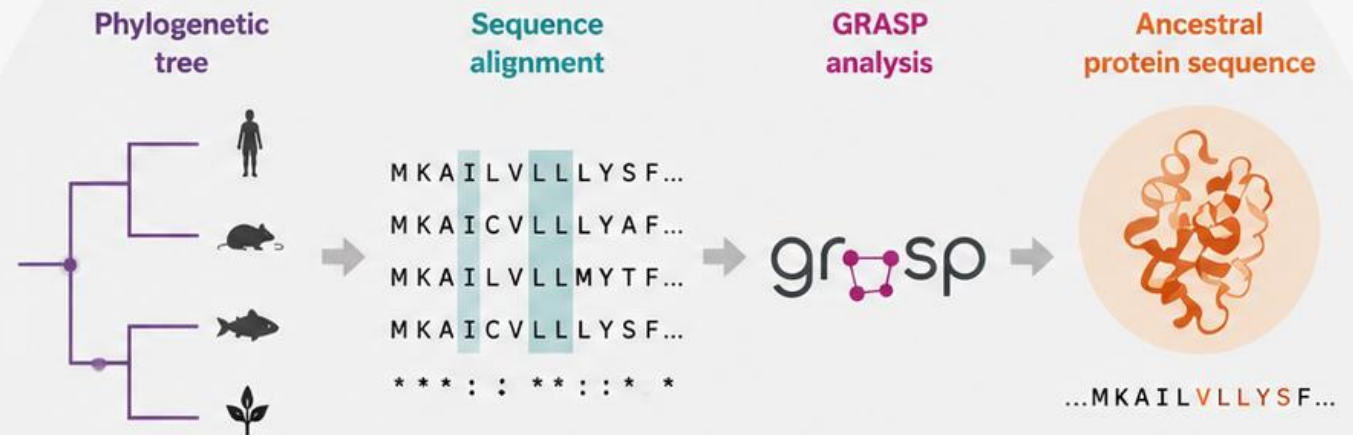
SCALES TO LARGE DATASETS

Designed to analyze very large alignments and phylogenetic trees efficiently.



EXPERIMENTALLY VALIDATED

Predicted ancestral proteins can be synthesized and tested in the laboratory.



GRASP has been successfully used to reconstruct ancient enzymes, including cytochrome P450 proteins, that were synthesized and shown to be functional in the laboratory.

Ancestral Sequence Reconstruction (GRASP / RStudio)

Since the GRASP web server was unavailable, ancestral sequence reconstruction (ASR) was performed in RStudio using the *phangorn* package.



1. INPUT DATA

- UndB multiple sequence alignment (124 proteins)
- IQ-TREE phylogenetic tree



2. MODEL FITTING

The alignment was fitted to the tree using a maximum likelihood substitution model.



3. MODEL OPTIMIZATION

Model parameters (edge lengths, invariant sites, gamma) were optimized to improve the likelihood.



4. ANCESTRAL SEQUENCE RECONSTRUCTION

Maximum likelihood was used to infer the most probable amino acid at each position for internal nodes.

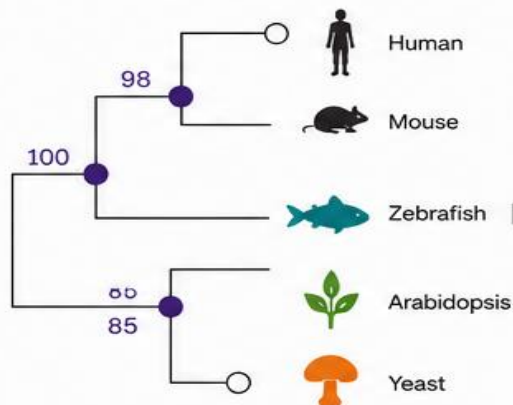


5. EXPORT RESULTS

Reconstructed ancestral sequences were exported as a FASTA file.

EXAMPLE WORKFLOW

1. PHYLOGENETIC TREE (IQ-TREE)



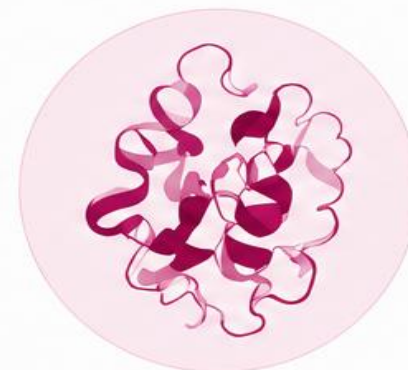
- Internal node (ancestral sequence)
- Observed tip (sequence from a species)

2. ANCESTRAL SEQUENCE RECONSTRUCTION (RStudio) (phangorn)

...MKA I L V L L L Y S F...
...MKA I C V L L L Y A F...
...MKA I L V L L M Y T F...
...MKA I C V L L L Y S F...
* * * : : * * * : : * *

Predicted ancestral amino acid states at the internal nodes.

3. ANCESTRAL PROTEIN SEQUENCES (ANALYSIS)



Reconstructed ancestral proteins can be modeled, analyzed and experimentally validated.



RESULT

Ancestral amino acid states were inferred for internal nodes of the UndB phylogenetic tree. These reconstructed sequences represent hypothetical ancestral UndB proteins.

Conclusion

IQ-TREE and GRASP together provide a powerful framework for **phylogenetic inference** and **ancestral protein reconstruction**.



KEY TAKEAWAY

Combining accurate tree inference with ancestral reconstruction enables us to explore the evolution of proteins and uncover their functional history.



ACCURATE PHYLOGENETIC INFERENCE

IQ-TREE reconstructs reliable phylogenetic trees using the Maximum Likelihood approach.



ANCESTRAL SEQUENCE RECONSTRUCTION

GRASP infers ancestral protein sequences at internal nodes of the tree.



FUNCTIONAL & EVOLUTIONARY INSIGHTS

Reconstructed proteins help us understand how functions evolved and how proteins adapted over time.



REAL-WORLD IMPACT

Predicted ancestral proteins can be synthesized and experimentally validated in the laboratory.



Together, IQ-TREE and GRASP open the door to deeper understanding of protein evolution and the molecular history of life.